

RFR Regime

(6. Single Curve Construction II)

Convexity Adjustment

- ▶ The Future rate and an equivalent forward rate are not the same. The Future rate is higher than the forward rate. The differential between Future Rate and Forward Rate is called the Convexity Adjustment.

$$\text{Future rate} - \text{Forward rate} = \text{Conv. Adj}$$

- ▶ The convexity appears because Futures are daily settled whereas FRAs are settled once at the time of settlement. The PnL on a future contract interacts with the prevailing rates to give rise to convexity.
- ▶ The Future rate : $R = 100 - P$, where P is the future price. The Rate and the price are inversely proportional. From a Future buyer perspective,

If $R \uparrow$, $P \downarrow$, it leads to losses. Money needs to be borrowed at higher rate to be deposited to Margin account. Similarly, if $R \downarrow$, $P \uparrow$, it leads to profit. But money gets credited to margin account at lower rate. In both cases, the buyer is at a disadvantage. So, buyer will demand lower price, which inflates the Future rate.

Convexity Adjustment

- ▶ There are both model based and model free formulas for convexity adjustments. The convexity adjustment can be shown as a byproduct of improper measure and timing of cashflows. (More on this later!). Under HW1F model,

- ▶ The convexity adjustment for a CME 1M SOFR future is given as

$$\left[T - \frac{1}{12}, T\right] \quad C^{1m}(0: T - \delta, T) = \frac{\sigma^2}{6} (3T^2 - 3T\delta + \delta^2) \quad \delta \approx 1/12, \sigma \approx 1\%$$

- ▶ The convexity adjustment for a CME 3M SOFR future $[T_{j-1}, T_j]$ is given as

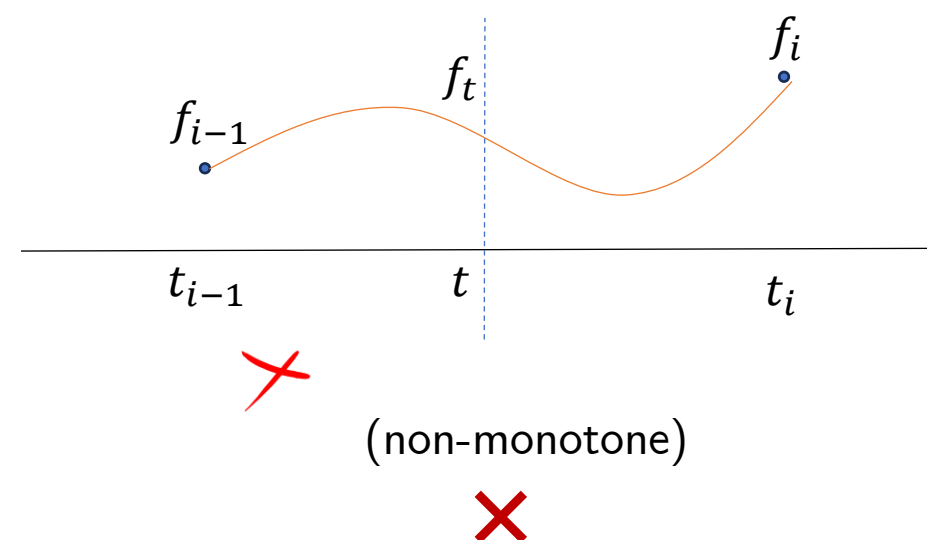
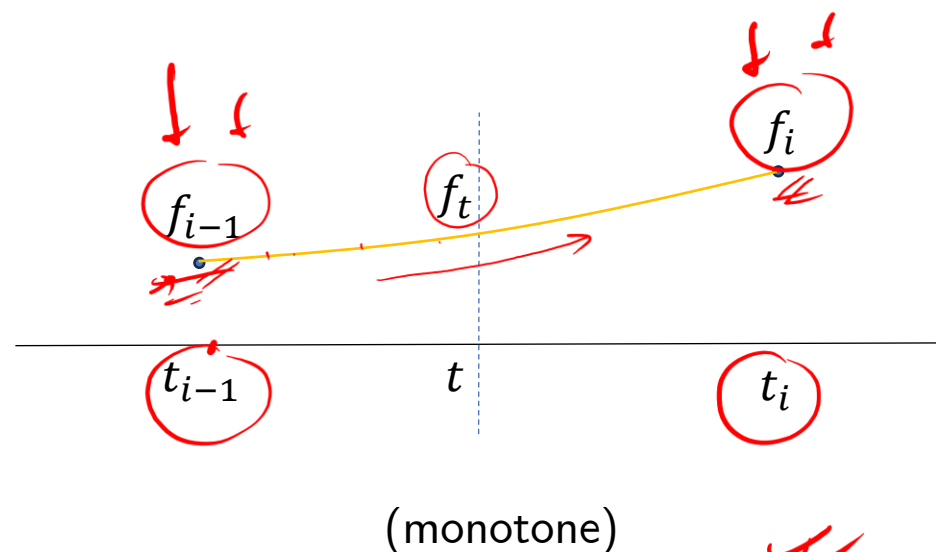
$$(T_{j-1}, T_j) \quad C^{3m}(0: T_{j-1}, T_j) = \frac{\sigma^2}{6} (2T_j^3 - 3T_j T_{j-1}^2 + T_{j-1}^3) \approx \frac{\sigma^2}{2} T_j^2$$

- ▶ For futures involving partially fixed daily rates, the convexity adjustment is prorated.



Monotone

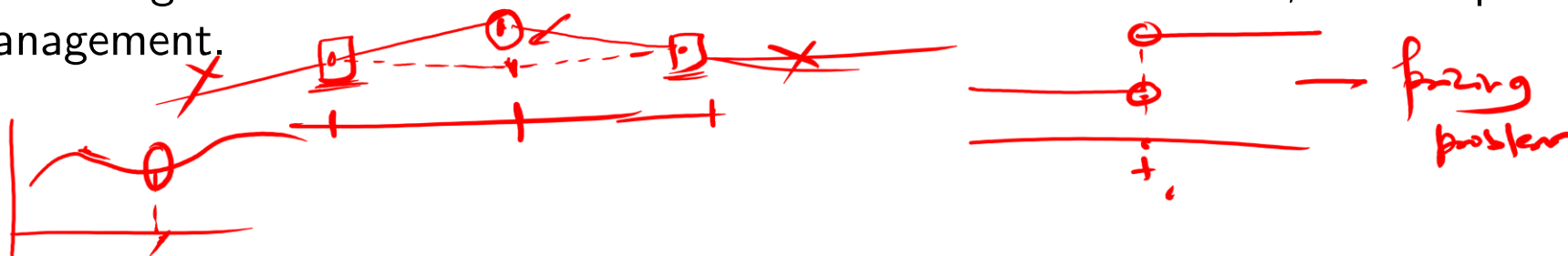
- It is important that interpolation schemes are **monotone**. Between, pairs of quotes, the interpolated forward rate must be strictly increasing or decreasing (no turns). This is to eliminate forward rate outliers which create arbitrage opportunities.



Global vs Local

- ▶ Forward rates derived from piecewise constant or linear interpolation only depend on adjacent nodes. These are called local methods. They have advantage of localized risk in the sense that any perturbation to input(quote) leads to a localized bump which is easier to manage and hedge.
- ▶ Splines on the other hand produce smooth forward curves which capture the market dynamics better. However, the interpolated rates depend on placement of wide range of knots. These are called global methods which produce risk that is more dispersed/non-local and is difficult to hedge.

Linear interpolation methods give rise to forward rate jumps which makes Forward rates undefined at certain nodes. This poses a direct challenge for derivative pricing. However, risk management becomes easy as the shocks are local. On the other hand, splines are attractive for pricing but a nightmare for risk management. It's not uncommon to maintain two risk curves, one for pricing and one for risk management.



Monotone Convex

- ▶ Monotone convex is a specialized implementation popularized by PATRICK S. HAGAN & GRAEME WEST(2006). The state variables are forward rates.
- ▶ The procedure ensures the interpolated rates are monotone and convex if the inputs have analogous properties. It also ensures forward rates are positive if the discrete rates are positive. The high-level steps include the following

❑ Obtain the discrete forward rates $\underline{f_i^d} = \frac{r_i \tau_i - r_{i-1} \tau_{i-1}}{\tau_i - \tau_{i-1}}$. This is viewed as the rate between the whole interval $[\tau_{i-1}, \tau_i]$

❑ Instantaneous forward rates at time τ_i is given as $\underline{f_i} = \frac{\tau_i - \tau_{i-1}}{\tau_{i+1} - \tau_{i-1}} \underline{f_{i+1}^d} + \frac{\tau_{i+1} - \tau_i}{\tau_{i+1} - \tau_{i-1}} \underline{f_i^d}$ for $i = 1, 2, \dots, n-1$

❑ The rates at boundary points τ_0, τ_n are selected so that $\underline{f'(\tau_0)} = \underline{f'(\tau_n)} = 0$. They are given as



$$\underline{f_0} = f_1^d - \frac{1}{2}(f_1 - f_1^d)$$

$$\underline{f_n} = f_n^d - \frac{1}{2}(f_{n-1} - f_n^d)$$

Monotone Convex

► An interpolator function is chosen to satisfy the following

□ $\frac{1}{(\tau_i - \tau_{i-1})} \int_{\tau_{i-1}}^{\tau_i} f(t) dt = f_i^d$, the discrete forward can be recovered from the curve

□ f is positive ✓

□ f is continuous ✓

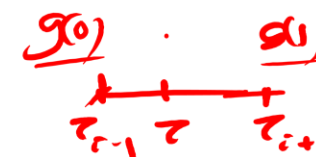
□ f is monotonous and convex

$$f_i^d \times (\tau_i - \tau_{i-1}) = \int_{\tau_{i-1}}^{\tau_i} f(s) ds$$



The interpolator is a local quadratic function between $[\tau_{i-1}, \tau_i]$ and is defined as $g(x) = K + Lx + Mx^2$

$$f(\tau) = g\left(\frac{\tau - \tau_{i-1}}{\tau_i - \tau_{i-1}}\right) + f_i^d$$



$$g(0)$$

For more details refer to the paper.

Minimalist quadratic interpolator - Splines

- We seek an interpolator of the instantaneous forward rates of the form

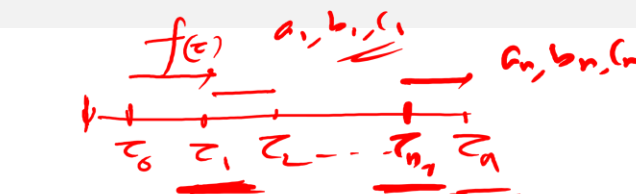
$$f(\tau) = a_i + b_i(\tau - \tau_{i-1}) + c_i(\tau - \tau_{i-1})^2 \quad \tau_{i-1} \leq \tau \leq \tau_i$$

- One required condition is

$$\frac{1}{\tau_i - \tau_{i-1}} \int_{\tau_{i-1}}^{\tau_i} f(t) dt = f_i^d$$

- Let $h_i = \tau_i - \tau_{i-1}$. We get $f_i^d = a_i + \frac{1}{2}b_i h_i + \frac{1}{3}c_i h_i^2$ $1 \leq i \leq n$

- Interpolant must be continuous. So, it leads to $a_{i+1} = a_i + b_i h_i + c_i h_i^2$



$$1 \leq i \leq n$$

$$1 \leq i \leq n - 1$$

This leads to a total $2n - 1$ linear equations and $3n$ unknowns

Minimalist quadratic interpolator – Penalty Function

- ▶ Instead of the typical constraints for slope and curvature continuity, we are going to use an error function with a regularization term to penalize sum of squared jumps in the first derivatives across boundaries and sum of squared second derivatives.

$$f'(\tau) = b_i + 2c_i(\tau - \tau_{i-1})$$

$$f''(\tau) = 2c_i$$

$2c_i$

- ▶ The jump in the first derivative is given as $J_{1,i} = b_{i+1} - b_i - 2c_i h_i$ $1 \leq i \leq n-1$

- ▶ We weight the second derivative by the interval to capture the total concavity. It's given as

$$J_{2,i} = (2c_i)h_i \quad 1 \leq i \leq n$$



Minimalist quadratic interpolator – Optimization

► Our final constrained optimization takes the form

$$\min_{(a,b,c)} \lambda \sum_{i=1}^{n-1} J_{1,i}^2 + (1 - \lambda) \sum_{i=1}^n J_{2,i}^2$$

Ⓢ → hyperparameter

s. t.

$$\rightarrow f_i^d = a_i + \frac{1}{2} b_i h_i + \frac{1}{3} c_i h_i^2 \quad 1 \leq i \leq n$$

$$\rightarrow a_{i+1} = a_i + b_i h_i + c_i h_i^2 \quad 1 \leq i \leq n - 1$$